

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1140

Orchard Road 138 kV
October 23, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1140. The Interconnection Customer proposes a 20.2 MW battery energy storage facility interconnecting at the Orchard Road 138 kV substation in the Pennsylvania area. The request was received on January 21, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 331 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,379,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Walnut Crk-Volcano 138.0 kV	Ft Humbug-Coronet 500.0 kV	242.3	240	12.5
Talenenergy Martins Crk Llc Lock Haven-El Cerrito Sta G 138.0 kV	Scott-American Crystal Sugar Drayton 345.0 kV L	122.5	235	5.1
Cayucos 345/1 kV Autotransformer	Sunrise Center-Lutesville 345.0 kV	112.1	675	9.8
Taylor Flats-Talenenergy Martins Crk Llc Lock Haven 138.0 kV	American Crystal Sugar Drayton-Scott Substation 345.0 kV	108.6	265	5.1
Cayucos Substation 500 kV/1 kV Transformer Bank	Sunrise Ctr-Lutesville 345.0 kV	108.1	700	9.8
Blackrock-Eklutna 345.0 kV	Caldwell-Piney Crk Project 345.0 kV	102.6	1,380	9.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Walnut Crk-Volcano 138.0 kV	Rebuild line with higher-capacity conductor	\$4,614,000	100% (\$4,614,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Talenenergy Martins Crk Llc Lock Haven-El Cerrito Sta G 138.0 kV	Rebuild line with higher-capacity conductor	\$9,432,000	100% (\$9,432,000)
Cayucos 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$13,872,000	2% (\$302,000)
Talenenergy Martins Crk Llc Lock Haven Substation-Taylor Flats 138.0 kV	Reconductor line	\$3,037,000	100% (\$3,037,000)
Cayucos Substation 500 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$34,491,000	2% (\$750,000)
Blackrock-Eklutna 345.0 kV Line	Reconductor line	\$23,734,000	1% (\$244,000)
	Total allocated to project		\$18,379,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.